Supplementary Material for *FreqDuet: Online Frequency-to-Authority Allocation for Hierarchical Bus Control*

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S1. Frozen evidence and decision ledger

The current paper package binds the gate-positive V8 decision, failed V9 long-training gate, and no-pass V33 development screen. The stages use disjoint training and evaluation seeds and are not pooled. The V8 decision label records its preregistered effect/no-harm gate; it does not imply a familywise-significant result at 0.05. V33 was not a confirmation stage and is listed separately from the confirmatory decision ledger.

Phase	Train x eval	Pairs	Registered decision	Claim eligible
V8 (40 ep)	6 x 4	24	Gate passed	Yes
V9 (200 ep)	8 x 8	64	Not confirmed	No

The term `confirmed_main` identifies the V8-selected configuration; it does not mean that V9 passed. The machine-readable package remains `submission_ready: false` with blocker `v9_longtrain_not_confirmed`.

S2. Exact current-controller configuration

The current controller resolves through the following inheritance chain:

1. `config_v2.yaml`
2. `configs_freqduet/F_freqduet_harmonic_hiro.yaml`
3. `configs_freqduet/F_freqduet_timetable_hiro.yaml`
4. `configs_freqduet/F_freqduet_terminal_hiro.yaml`
5. `configs_freqduet/F_freqduet_terminal_waitattr_hiro.yaml`
6. `configs_freqduet/F_freqduet_terminal_spline2dir_waitattr_hiro.yaml`
7. `configs_freqduet/F_freqduet_terminal_aligned_nopromotion_hiro.yaml`
8. `configs_freqduet/F_freqduet_terminal_lowerhf_poswait_hiro.yaml`
9. `configs_freqduet/F_freqduet_terminal_lowerhf_localcap10_w06_hiro.yaml`
10. `configs_freqduet/F_freqduet_terminal_promotion_localcap10_w06_hiro.yaml`
11. `configs_freqduet/F_freqduet_terminal_energydenoise_hold45_hiro.yaml`
12. `configs_freqduet/F_freqduet_terminal_energydenoise_hold45_holdpen03_hiro.yaml`
13. `configs_freqduet/F_freqduet_terminal_energydenoise_hold45_holdpen03_strictprom_hiro.yaml`
14. `configs_freqduet/F_freqduet_terminal_energydenoise_hold45_holdpen03_adaptstrict_hiro.yaml`
15. `configs_freqduet/F_freqduet_terminal_energydenoise_hold45_holdpen03_adaptstrict_local_hiro.yaml`
16. `configs_freqduet/F_freqduet_protocol_v2_main_hiro.yaml`
17. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_hiro.yaml`
18. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_markov_hiro.yaml`
19. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_markov_smdp_hiro.yaml`
20. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_rebuild_nophys_hiro.yaml`
21. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_rebuild_hiro.yaml`
22. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_rebuild_strict_hiro.yaml`
23. `configs_freqduet/F_freqduet_protocol_v2_uppercompact_strict_intervaladd_hiro.yaml`

24. `configs_freqduet/F_freqduet_protocol_v3_compact_b30_hiro.yaml`
25. `configs_freqduet/F_freqduet_protocol_v4_main_hiro.yaml`
26. `configs_freqduet/F_freqduet_protocol_v5_main_hiro.yaml`
27. `configs_freqduet/F_freqduet_protocol_v6_main_hiro.yaml`
28. `configs_freqduet/F_freqduet_protocol_v6_noguard_hiro.yaml`
29. `configs_freqduet/F_freqduet_protocol_v6_avlcompact_hiro.yaml`
30. `configs_freqduet/F_freqduet_protocol_v6_avlcompact_w2_hiro.yaml`
31. `configs_freqduet/F_freqduet_protocol_v6_confirmed_main_hiro.yaml`

Key resolved settings are: harmonic historical prior; 60-s causal bins; four Fourier harmonics over 14 h; 30-min low-frequency forecast; 15-min upper replanning over 45 min; rolling zero-sum V6 headway budget; executable terminal dispatch; low-frequency upper and high-frequency lower authority; compact same-time APC/AVL lower context; previous-action state; holding actions `{0, 5, 10, 15, 20, 30, 45}` s; and a weight-two pre-action two-sided regularity reward. The legacy holding guard, promotion, and leakage penalty are disabled.

S3. Causal and physical contract

Passenger arrivals become visible only after their within-bin arrival times. APC counts enter the frequency tracker only when a complete 60-s bin closes. The lower regularity tuple freezes the matched predecessor departure, same-time follower AVL estimate, target headway, and action before transition settlement. The categorical action is executed without post-policy clipping. Commanded and realized holding are recorded separately. The upper timetable materializes future launch times once; cached reuse is read-only; each closed rolling budget block conserves its headway adjustment; and actual launch cannot precede either vehicle readiness or the executable scheduled time.

S4. Complete V8 and V9 policy-reference outcomes

Outcome	V8 delta [95% CI]	V8 Holm p	V9 delta [95% CI]	V9 Holm p
Restricted passenger journey (min)	-0.263 [-0.837, +0.175]	0.938	-1.242 [-2.204, -0.536]	0.047
Restricted passenger wait (min)	-0.185 [-0.470, +0.050]	0.375	-1.011 [-1.878, -0.397]	0.047
Restricted in-vehicle time (min)	-0.078 [-0.376, +0.147]	1.000	-0.232 [-0.355, -0.121]	0.047
Headway coefficient of variation	-0.022 [-0.038, -0.008]	0.125	-0.009 [-0.028, +0.006]	0.031
Unserved passengers (percentage points)	+0.02 [-0.00, +0.11]	1.000	+0.00 [+0.00, +0.02]	0.250
Realized holding (s/launched trip)	-1.7 [-41.8, +29.4]	1.000	-37.6 [-59.8, -16.2]	0.070
Trips denied at least once (percentage points)	+0.48 [-9.21, +8.81]	1.000	-1.65 [-3.96, -0.22]	0.117
Restricted service cost	-0.040 [-0.079, -0.007]	0.125	-0.110 [-0.207, -0.042]	0.047

All entries are current policy minus `F_freqduet_protocol_v6_noguard_hiro`. The V8 row corresponds to the source config `F_freqduet_protocol_v6_avlcompact_w2_hiro` and the V9 row to its exact paper alias `F_freqduet_protocol_v6_confirmed_main_hiro`. Both arms disable the legacy holding guard. The contrast combines compact APC/AVL context with the incremental regularity objective.

S5. V33 lower-actor timescale development screen

V33 tested whether stopping lower-actor updates after episode 40, 80, or 100 could preserve the short-training regularity effect at checkpoint 199. The lower critics and both upper components continued training. Each candidate used four training seeds crossed with four evaluation seeds. Values below are candidate minus the Protocol V6 reference; lower is better.

Lower actor frozen after episode	Headway CV delta [95% CI]	Negative train seeds	Journey delta [95% CI], min	Gate passed
100	-0.007 [-0.049, +0.030]	2/4	-1.050 [-2.369, -0.302]	No
80	-0.004 [-0.031, +0.015]	2/4	-1.491 [-2.937, -0.491]	No
40	-0.012 [-0.038, +0.008]	4/4	-1.954 [-3.190, -1.077]	No

All three candidates met the journey no-harm condition, but none achieved a headway-CV reduction of at least 0.02 with its interval below zero. The episode-40 schedule had a negative headway-CV difference for all four training seeds, but its mean reduction was 0.012 and its interval crossed zero. V33 therefore returned `no_pass`. The confirmation roster was not used. The unfrozen current controller passed the incremental gate on this development roster, but it was a control rather than a registered V33 candidate. It cannot replace the failed independent V9 result.

S6. Complete V9 external-baseline outcomes

Baseline	Outcome	FreqDuet mean	Baseline mean	Delta [95% CI]	Holm p
Fixed headway	Restricted passenger journey (min)	+20.349	+17.879	+2.470 [+1.874, +3.188]	0.023
Fixed headway	Restricted passenger wait (min)	+8.009	+7.001	+1.009 [+0.439, +1.704]	0.023
Fixed headway	Restricted in-vehicle time (min)	+12.340	+10.878	+1.461 [+1.367, +1.542]	0.023
Fixed headway	Headway coefficient of variation	+0.213	+0.432	-0.219 [-0.238, -0.198]	0.023
Fixed headway	Unserved passengers (percentage points)	+0.14	+0.21	-0.07 [-0.20, -0.00]	0.023
Fixed headway	Realized holding (s/launched trip)	+286.8	+0.0	+286.8 [+269.7, +302.0]	0.023
Fixed headway	Trips denied at least once (percentage points)	+85.85	+21.76	+64.10 [+61.97, +66.03]	0.023
Fixed headway	Restricted service cost	+1.021	+1.143	-0.122 [-0.180, -0.050]	0.023
Rule holding	Restricted passenger journey (min)	+20.349	+23.573	-3.224 [-4.727, -1.885]	0.023
Rule holding	Restricted passenger wait (min)	+8.009	+10.799	-2.790 [-4.208, -1.519]	0.023
Rule holding	Restricted in-vehicle time (min)	+12.340	+12.774	-0.434 [-0.544, -0.335]	0.023
Rule holding	Headway coefficient of variation	+0.213	+0.286	-0.073 [-0.082, -0.062]	0.023
Rule holding	Unserved passengers (percentage points)	+0.14	+0.13	+0.01 [+0.00, +0.03]	0.023
Rule holding	Realized holding (s/launched trip)	+286.8	+360.2	-73.3 [-91.2, -57.3]	0.023

Baseline	Outcome	FreqDuet mean	Baseline mean	Delta [95% CI]	Holm p
Rule holding	Trips denied at least once (percentage points)	+85.85	+89.03	-3.17 [-5.42, -1.90]	0.023
Rule holding	Restricted service cost	+1.021	+1.372	-0.351 [-0.496, -0.221]	0.023
Rule MPC	Restricted passenger journey (min)	+20.349	+46.860	-26.511 [-31.003, -21.628]	0.023
Rule MPC	Restricted passenger wait (min)	+8.009	+33.527	-25.517 [-29.945, -20.699]	0.023
Rule MPC	Restricted in-vehicle time (min)	+12.340	+13.333	-0.994 [-1.116, -0.881]	0.023
Rule MPC	Headway coefficient of variation	+0.213	+0.207	+0.006 [-0.006, +0.020]	0.023
Rule MPC	Unserved passengers (percentage points)	+0.14	+0.05	+0.09 [+0.00, +0.27]	0.023
Rule MPC	Realized holding (s/launched trip)	+286.8	+409.6	-122.7 [-139.7, -107.3]	0.023
Rule MPC	Trips denied at least once (percentage points)	+85.85	+24.33	+61.52 [+59.09, +63.77]	0.023
Rule MPC	Restricted service cost	+1.021	+3.562	-2.541 [-2.984, -2.060]	0.023

Means and differences use the same outcome units shown in each row. The denied-trip rate is the fraction of scheduled trips denied at least once by the fixed-pool readiness check; retry duration is reported separately in the source artifacts. A launch rate of one does not imply a zero denial rate.

S7. Statistical procedure

Each train-seed policy is evaluated on every registered evaluation seed under common random numbers. Crossed-bootstrap intervals resample training seeds and a shared set of evaluation seeds, preserving policy pairing. Sign-flip tests operate on train-seed mean deltas and therefore target conditional training-seed inference rather than the crossed population represented by the bootstrap. Holm adjustment is performed across compared methods separately for each outcome. V8, V9, and V33 estimates are not pooled, and no failed gate is rescued by a favorable secondary endpoint.

The V8 gate required a headway-CV improvement of at least 0.02 against the Protocol V6 reference, at least 0.01 against compact context alone, passenger journey within +0.15 min of both references, complete paired rollouts, at least 50% same-time follower coverage, zero execution adjustment, and preservation of holding and denied-dispatch gains. V9 reused those gates and additionally required a headway-CV interval below zero, a journey CI upper bound no larger than +0.15 min, and negative train-seed CV differences for at least 75% of training seeds. V9 met the latter two added directional/no-harm conditions but failed the inherited CV magnitude gates and CI-exclusion requirement.

V33 reused the V8 effect and mechanism thresholds. It additionally required a headway-CV interval below zero, a journey CI upper bound no larger than +0.15 min, and negative train-seed CV differences for at least 75% of training seeds. A development candidate had to pass every condition before the reserved confirmation seeds could be used. None did.

S8. Frequency-claim boundary and negative development evidence

The current confirmatory package does not contain a same-stage NoFreq, RawHistory, AllFreq, swapped-layer, promotion, or leakage ablation. Earlier V6 engineering screens used some of these controls, but they were exploratory, incomplete, or tied to superseded controller semantics. They cannot be used as confirmatory evidence that frequency separation itself caused the V8 effect. Likewise, V28-V32 counterfactual value/rank/margin planners and all three V33 lower-actor freeze schedules failed their registered development gates and were not promoted. These negative results define the stopping decision: the manuscript uses the frozen V6 controller and does not tune another gate on the same development contexts.

S9. External-data provenance

Figure 5 uses separately normalized demand shapes from the local FreqDuet OD input, 39 complete station-complex days selected from a bounded public MTA AFC cache, and seven complete routes spanning 37 route-days selected from a bounded Halifax APC cache. Incomplete pagination fragments were excluded before aggregation. MTA Bus Time is used only as route/stop and AVL audit data, not APC or onboard load. MBTA boarding, alighting, and load data provide separate calibration targets but are not a same-day matched calibration of the simulated network. API credentials are not stored in the repository or paper package.

S10. Reproducibility boundary

The paper evidence directory tracks the small V8, V9, and V33 CSV and JSON artifacts, exact seed contracts, source commits, and resolved config lineage. Checkpoints and full training logs are intentionally not part of the manuscript bundle. A build validates the frozen package before rendering any text or table. A successful build verifies consistency; it does not change the scientific decision or create a field-deployment claim.